

POTHOLELOG · DESIGN DECISIONS REGISTER

PotholeLog design decisions register

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DOCUMENT CONTROL

Document ID	PHL-DEC-001	Version	0.5
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potholelog		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	Amish accepted the cost overrun against the value-engineering target on 2026-10-03; row added to decisions made; value engineering section updated
0.4	2026-10-02	Amish Chadha	Value engineering: the light pipes and LEDs, breather vent and nameplate label are now BOM lines 14 to 16; estimate USD 77.50, USD 2.50 over the target
0.3	2026-10-02	Amish Chadha	Design for construction (PHL-DDR-003, Tables 1 and 2) accepted by Amish on 2026-10-02; the 2026-10-01 row no longer says open for review
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all nine open decisions on 2026-10-02 (PHL-DDR-003 A1 to A3 accepted); moved to decisions made; value engineering notes the effect of the light pipes and vent
0.1	2026-10-01	Amish Chadha	

Register opened with the build plan;
budget treated as a value-engineering
target

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in the review note; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The box: 120 x 90 x 55 mm outside, 2.5 mm walls, a flat floor and four moulded corner pillars about 10 mm across	The standoff pattern, carrier size and clearances are set by them	PHL-DDR-003, P1, P7
2	The converter's size (about 30 x 40 mm) and its four mounting holes; its 60 V input rating on the datasheet	The carrier's module holes are marked from it; safety stop S3	PHL-DDR-003, P2; PHL-DDR-002, N1
3	The ESP32-S3 board is no larger than about 52 x 26 mm and its microSD slot can be reached with the lid off	Many ESP32-S3 boards are longer; a longer board needs the layout rechecked	PHL-DDR-003, P3
4	The IMU breakout: about 20 x 20 mm, two mounting holes, a side-entry plug, and the noise figure the calculations assume	The floor and plate IMU holes and the 3.4 mm clearance under the carrier	PHL-DDR-003, P5; PHL-CAL-001
5	The GNSS module: no larger than 28 x 28 mm, a timing-pulse output, and a plug-in lead long enough to set the lid aside	The lid layout and the wiring	PHL-DDR-003, P6
6	The hex standoffs' male thread is 6 mm or shorter	A longer thread would stick out under the plate	PHL-DDR-003, P1
7	The M6 bolt length for the vehicle's floor or crossmember	It depends on the structure at the chosen fixing points	PHL-PRC-001

■ Value engineering

Value-engineering target: USD 75 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 77.50 (USD 2.50 over the target). Main cost drivers and savings worth trying:

Amish accepted this overrun on 2026-10-03: the estimated cost of USD 77.50 against the USD 75 target (USD 2.50 over), accepting the USD 2.50 overrun on R15. Amish: "Cost over target - i accept all the cost variations and overruns". It stays reported against the target as an accepted overrun, and the savings below remain worth trying.

- The largest lines are the GNSS module (USD 15), the controller (USD 10), the 60 V converter with its protection (USD 9) and the enclosure (USD 8); the plate, IMU and memory card are USD 6 each.
- Making the design constructable added the carrier plate (USD 2.00) and the box fixing kit (USD 1.50) and moved USD 0.50 of hardware out of line 11: from USD 71.00 to USD 74.00.

- Adopted on 2026-10-02 and now in the BOM: the two status light pipes with their LEDs and lead (line 14, USD 1.00), the breather vent (line 15, USD 2.00) and the nameplate label (line 16, USD 0.50). Together they take the estimate from USD 74.00 to USD 77.50, USD 2.50 over the target.
- Savings worth trying: cut the plate and carrier from offcuts; a 16 GB card still holds about 66 to 80 days of raw data against the 30 days needed (about USD 2 less), which with offcut plates would close most of the gap; GNSS modules of the same class vary widely in price; buying the standoffs, screws and lead parts in packs lowers the per-unit hardware cost.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D7: bicycles a later variant; depot Wi-Fi first; floor mount above the rear axle; ESP32-S3; LSM6DSO and M10 classes; 100 m segments in CSV and GeoJSON; publish road-level results only	Amish: "i accept all your recommendations, go with them across all repos."	PHL-DDR-001, PHL-DDR-002
2026-09-25	N1: a 60 V-rated converter, with budget_usd raised from \$70 to \$75	Amish, same instruction	PHL-DDR-002
2026-09-25	N2: R5 defect-detection range 10 to 30 km/h with repeat slow passes; 300 mm reference defect kept	Amish, same instruction	PHL-DDR-002
2026-10-01	Design for construction: hex standoffs, carrier plate, module layout, gland locknut, IMU screwed into the plate, GNSS on foam tape, modelled lid pillars	Made under Amish's 2026-09-30 instruction to make the design physically buildable ("fix the design assumptions to match and be physically feasible as you draw the illustrations"); the changes themselves were accepted on 2026-10-02 (below)	PHL-DDR-003
2026-10-02	First pilot partner (first candidate to approach): a Dallas-Fort Worth area city with its own bus or refuse fleet, with TxDOT pavement management roughness data on state roads in that city as the reference IRI sections	Amish: "i approve your recommendations for all 555 open decisions."	PHL-DDR-001, O1; PHL-DDR-002
2026-10-02	Clear lid window over the GNSS antenna shown in the renders only (option a); the build keeps the opaque lid	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 1
2026-10-02	Two sealed status light pipes in the lid, for power and logging, adopted (option a); the two lid holes are sealed like the others	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 2
2026-10-02	Pressure-equalizing breather vent on the front end wall adopted (option a)	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 3
2026-10-02	Printed nameplate label with a forward arrow adopted (option a)	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW 2026-09-26, item 4
2026-10-02	Segment files to CityTwin: CityTwin fetches the daily segment files from the operator's server (option a, matching CityTwin's outbound pull, CTW-DDR-001 D11); a date per segment may be published but never pass times, and only once at least two vehicles or several days are pooled	Amish: "i approve your recommendations for all 555 open decisions."	PHL-DDR-002, cross-repo actions
2026-10-02	Floor holes sealed with a ring of neutral-cure silicone round each hole for the prototype (option a), spray-tested at TRL 4 before a	Amish: "i approve your recommendations for all 555 open decisions."	PHL-DDR-003, A1

Date	Decision	Decided by	Record
	pilot; the foam gasket is the fallback if the spray test leaks		
2026-10-02	GNSS module held by acrylic foam tape (option a), with adhesion checked after the TRL 4 heat soak; the printed clip is the fallback if the tape lets go	Amish: "i approve your recommendations for all 555 open decisions."	PHL-DDR-003, A2
2026-10-02	Keep the 160 mm plate and bolt the logger down as a unit with a ring spanner for the prototype (option a); review after a TRL 4 trial fitting	Amish: "i approve your recommendations for all 555 open decisions."	PHL-DDR-003, A3; PHL-CAL-001 [I4]
2026-10-02	Design for construction accepted: the changes in Tables 1 and 2 (P1 to P7 and their knock-on changes), as made	Amish: "APPROVED: Design-for-construction changes in 10 repos (CityTwin, CoolShade, PalletPilot, Heliolite, PotholeLog, EarthPress, ReadyKit, CellCheck, CargoMule and ThermaCart)"	PHL-DDR-003 , Tables 1 and 2
2026-10-03	Cost overrun accepted: the estimated cost of USD 77.50 against the USD 75 target (USD 2.50 over), accepting the USD 2.50 overrun on R15	Amish: "Cost over target - i accept all the cost variations and overruns"	REVIEW.md , session 2026-10-03